

THE CAREGIVER COGNITIVE-LOAD AGENT

Less to carry. More care to give.

A shared care picture. A clear next step.
An agent that helps while the caregiver stays in control.

Care Plan

Care Circle

Care Organizer

Care Handover

Grounded Agent

Frincy Clement · Maneetta Antony · Tejaswini Venkata Raju · Shrijan Chipalu

ONE CAREGIVER. TOO MUCH TO REMEMBER.

Everyone has part of the story. Maya holds it all together.

She coordinates Alex's care around work and family, remembering every change and resolving schedule conflicts again and again.

The invisible work of care

Keep the information in her head. Spot the clash. Find cover. Update everyone.

REMEMBER THE CHANGE

Physio moved
to 3:30

SCHEDULE CONFLICT

Work meeting
at 3:00

KEEP TRACK

Refill pickup
tomorrow

RESOLVE THE GAP

Who can drive
Alex?

Maya · daughter, caregiver, coordinator

Illustrative persona and scenario

THE PROBLEM

Care happens across people. The mental checklist stays with one.

“I thought you were taking Alex tomorrow.”

A change in one place can leave the ride, reminder, or caregiver assignment pointing to the old plan.

Every handover costs attention

The next caregiver needs current state, open risks, reliable contacts, and the exact items requiring review.

More reminders are not enough

The practical need is shared context, a named owner, a safe decision, and a visible outcome.

Enable Maya to coordinate the next step without rebuilding the whole picture.

MEET CARESTEAD

You care. We plan.

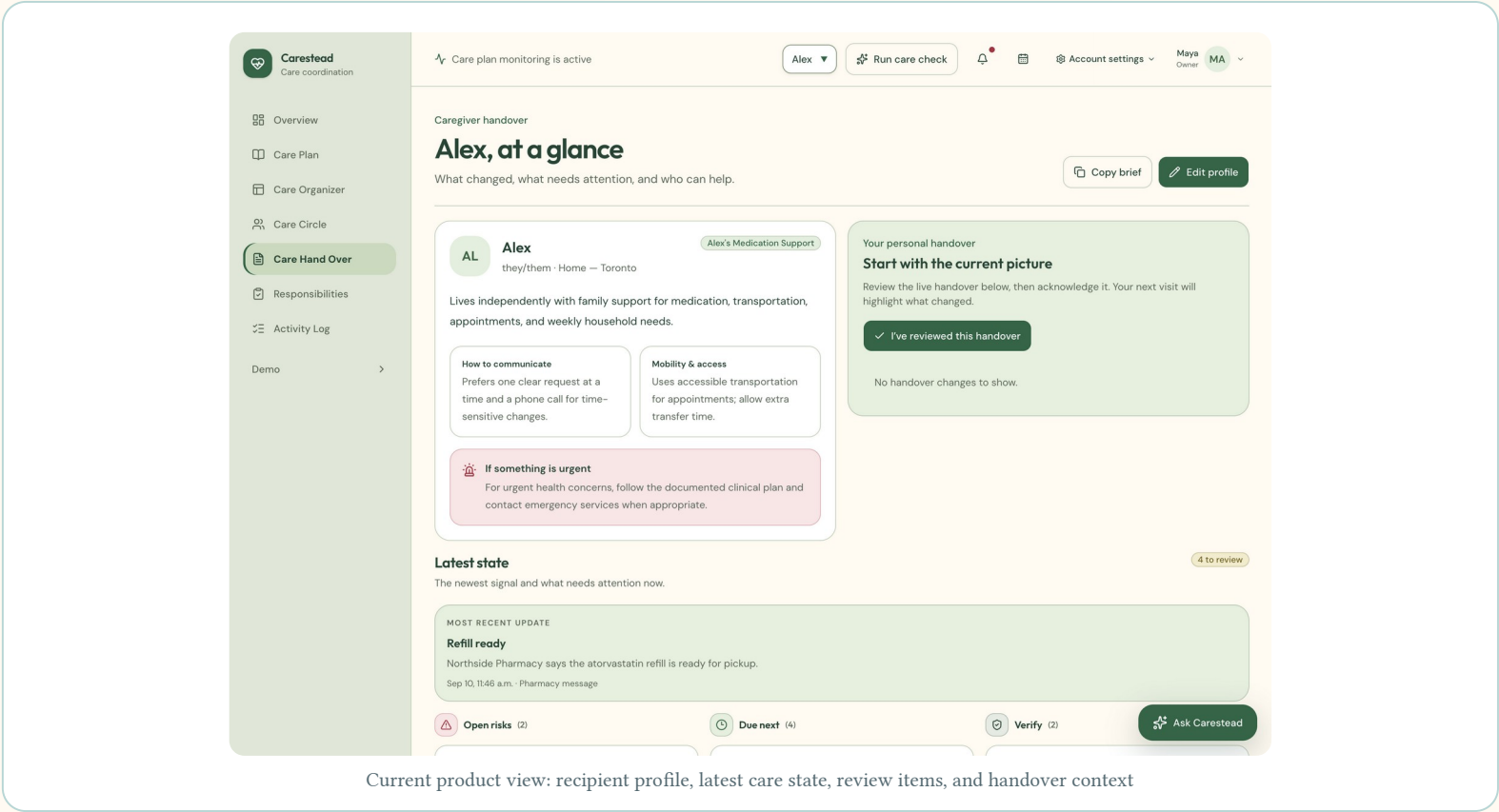
One place for the plan, people, evidence, and next safe step.

Care Plan
Reusable templates plus person-specific preferences, routines, goals, and supports.

Care Circle
Roles, contacts, invitations, responsibility owners, and trusted support.

Care Organizer
Plan breaks, activities, coverage, transport, and dependencies in one flow.

Care Handover
A concise live brief: current state, changes, risks, contacts, and watch items.



Current product view: recipient profile, latest care state, review items, and handover context

PRODUCT MAP

The complete Carestead workspace.

A person-centered care model connects context, coordination, people, actions, and governance.

Understand the person

- Overview + profile
- ✓ Care Plan + templates
- ✓ Preferences + routines
- ✓ Memory + current risks

Coordinate care

- Responsibilities
- ✓ Care Organizer
- ✓ Calendar scheduling
- ✓ Timeline + activity log

People + handover

- Care Circle + roles
- ✓ Contacts + support systems
- ✓ Live Handover
- ✓ Notifications

Agent + governance

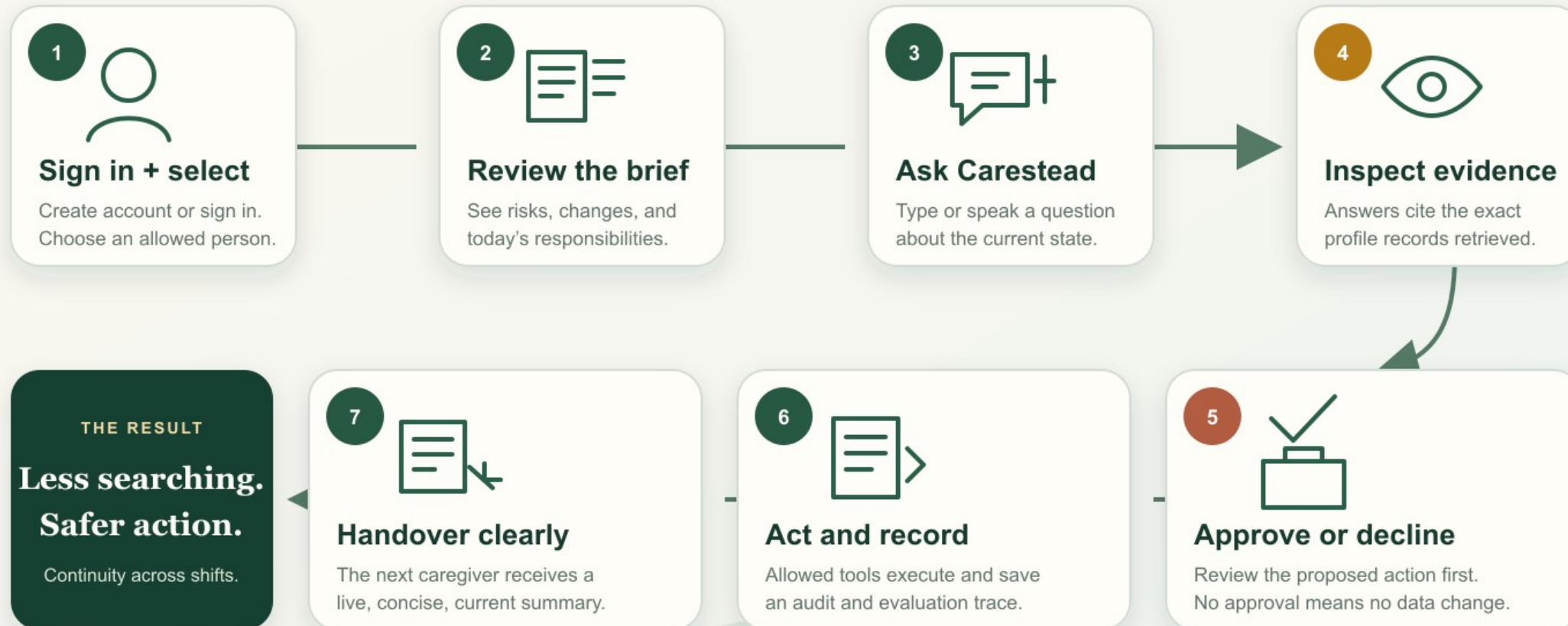
- Grounded chat + voice
- ✓ Document + image intake
- ✓ Integrations
- ✓ Consent + privacy + evals

Foundation: sign-in and roles · recipient scope · export and deletion · approval history · audit trail



A caregiver's journey

FROM SIGN-IN TO A SAFE HANDOVER



Every question, decision, approval, tool call, and outcome remains recipient-scoped and auditable.

AGENTIC DESIGN

Not every workflow uses AI the same way.

Carestead applies model intelligence selectively and keeps safety-critical control in deterministic code.

LLM-assisted workflow

MODEL	Extracts, summarizes, ranks, or drafts.
CONTROL LAYER	Validates, clarifies, scopes, and saves after review.

Full agentic workflow

MODEL	Interprets the goal and explains options.
CONTROL LAYER	Retrieves, plans, calls tools, requests approval, and verifies.

Grounded read-only

MODEL	Produces an evidence-backed answer or summary.
CONTROL LAYER	Retrieves recipient-scoped records and checks citations.

Deterministic agent

MODEL	Optional explanation only.
CONTROL LAYER	Rules detect risks, enforce policy, select tools, and trace outcomes.

Interaction modes: text · voice transcript · spoken reply · image · PDF · structured records

CAPABILITY MAP

Eight workflows. Four capability patterns.

Each workflow states what the model contributes and what the agent controls.

Grounded care chat

LLM: understand and synthesize

HYBRID AGENTIC

Agent: retrieve, cite, route, approve, execute

Adaptive care-plan builder

LLM: propose plan and missing details

LLM PLANNING

Agent: isolate, validate, approve, create tasks

Conflict resolution

LLM: rank and explain

FULL AGENTIC

Agent: simulate, validate, reschedule, verify

Communication composer

LLM: draft and adapt tone

LLM + DELIVERY

Agent: audience, channel, approval, receipt

Care-update extraction

LLM: extract intent and details

LLM ASSISTED

Agent: clarify, validate, match, approve, save

Document and image intake

LLM: extract fields and uncertainty

MULTIMODAL LLM

Agent: consent, validate, approve, track

Intelligent handover

LLM: evidence-backed summary

GROUNDING READ ONLY

Agent: scoped retrieval and citation checks

Risk checks and memory

LLM: optional explanation

DETERMINISTIC AGENT

Agent: rules, retrieval, policy, traces

PROTOTYPE USABILITY

Ask naturally. See the evidence. Approve the exact next step.

1 Grounded answer

The LLM answers from the selected person's current records and returns evidence IDs.

2 Action proposal

The orchestrator converts the request into a scoped, validated preview. It does not write immediately.

3 Approval + outcome

An authorized caregiver approves. Carestead executes once and records the result.

I prepared this action but have not changed anything yet. Review the details and approve it if they are correct.

▼ Evidence used · 2

PROPOSED TOOL

reschedule task

SAFETY POLICY

Explicit caregiver approval is required before any chat-initiated write.

Approval required

RESCHEDULE TASK

Move "Physiotherapy appointment" to Sep 14, 3:30 PM.

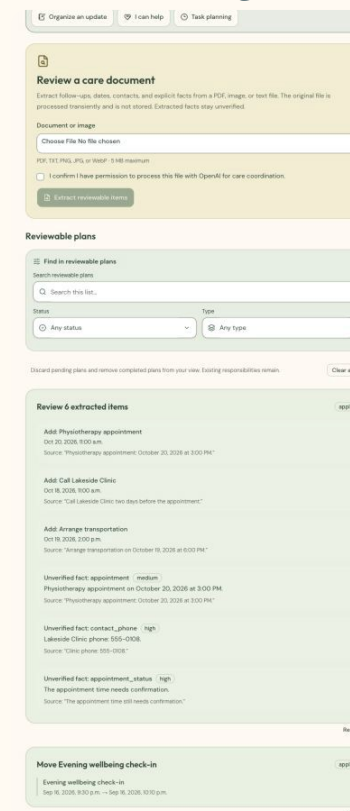
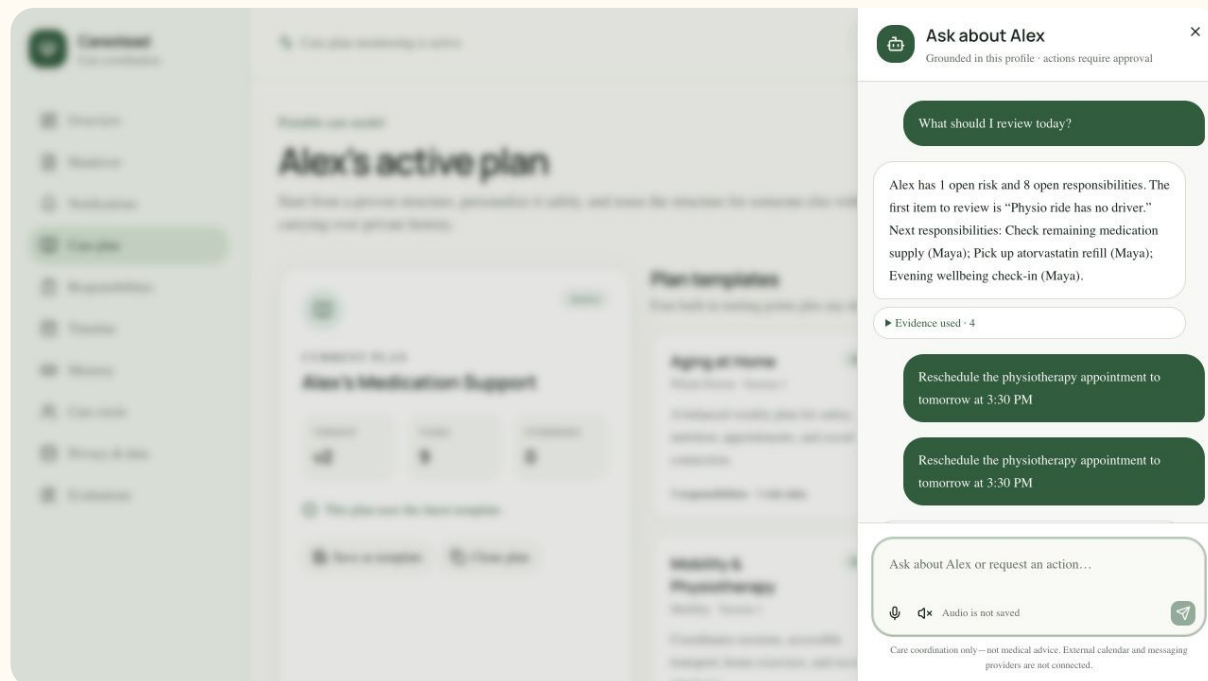
✓ Approve

✕ Cancel

VOICE + MULTIMODAL

Care updates arrive in the format caregivers already have.

Speak a request, type a note, or upload an image or PDF. Review structured changes before saving.



Voice interaction

Speech-to-text starts a grounded chat. Spoken replies are optional. Carestead stores transcript text, not raw microphone audio.

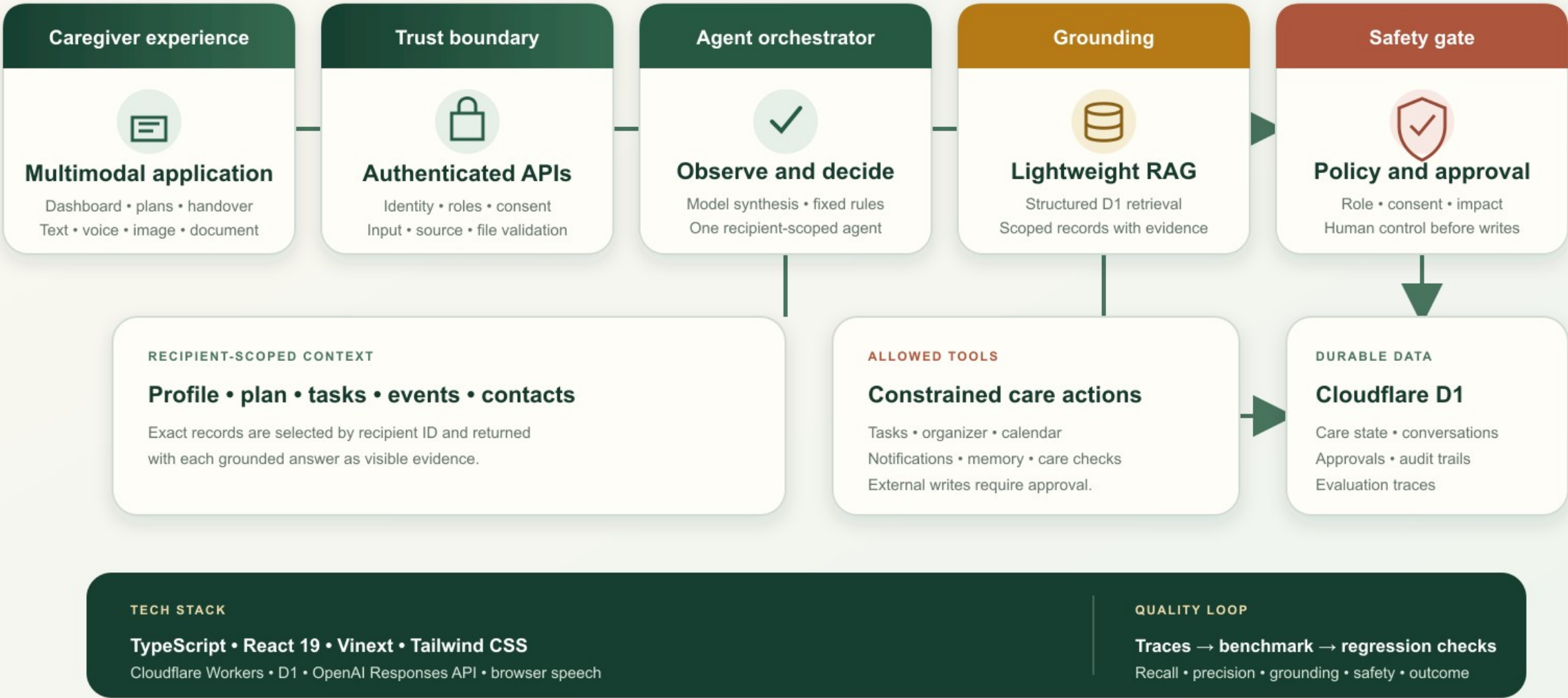
Multimodal extraction

The model extracts dates, contacts, follow-ups, facts, and uncertainty. The agent validates scope and asks for approval before saving.



Carestead technical architecture

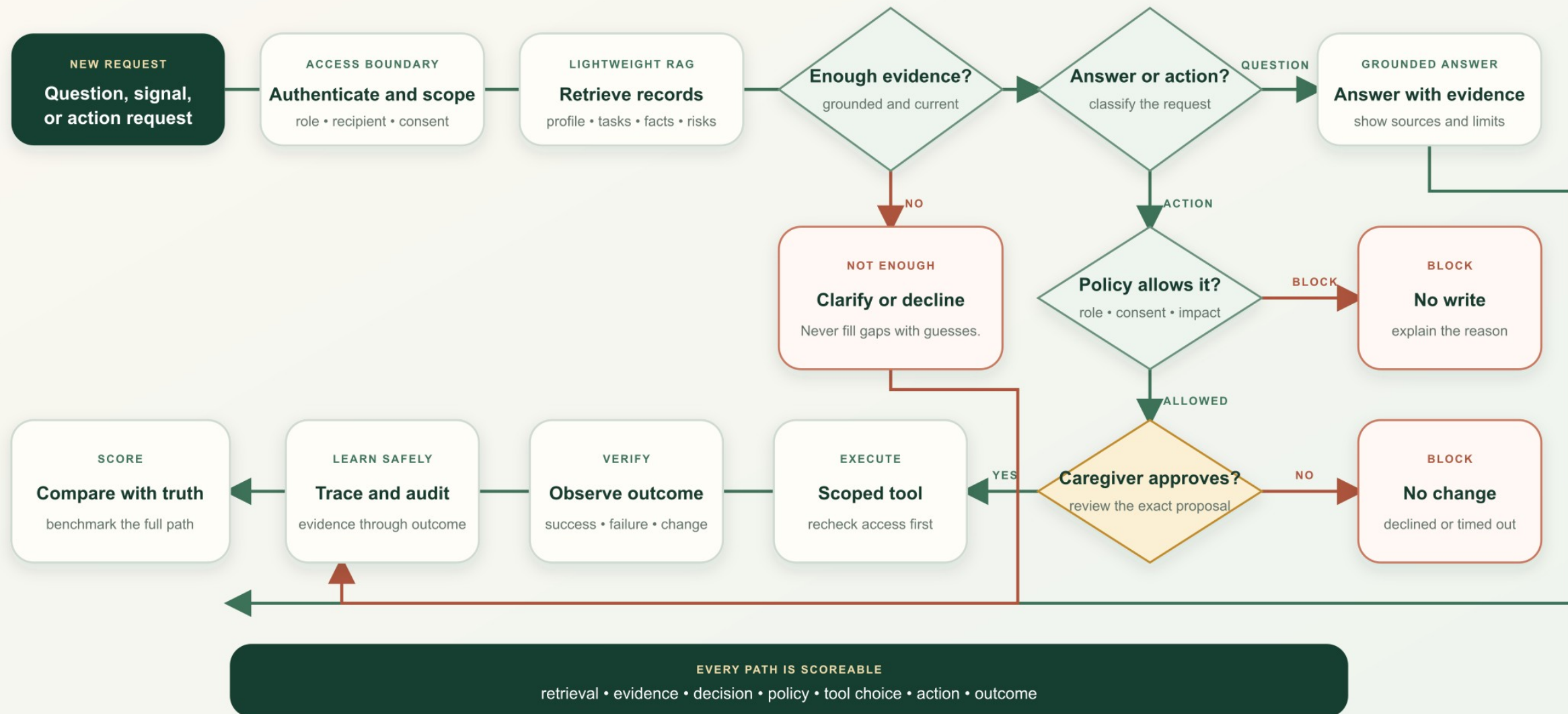
A SIMPLE, GROUNDED, APPROVAL-FIRST SYSTEM





Carestead decision path

GROUNDING ANSWERS • CONSTRAINED ACTIONS • HUMAN CONTROL



BOUNDED AUTONOMY

The system can reason broadly. It can act only through narrow, reviewable tools.

LLM contributes

Interpret intent · extract facts · summarize evidence · draft language · rank already-feasible options

Orchestrator decides

Recipient scope · tool routing · role checks · consent · risk rules · feasibility · idempotency

Caregiver decides

Approve, edit, or reject calendar changes, assignments, messages, plan changes, and other consequential writes

Profile ♦ plan

Responsibilities

Calendar

Care Circle

Handover

Notifications

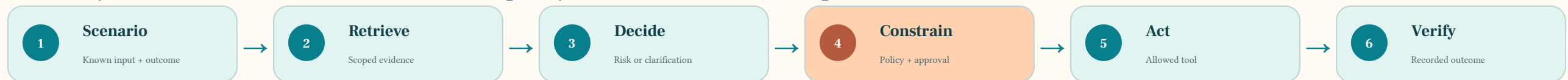
Memory

Audit ♦ evals

EVALUATION

Evaluate every step, not only the final answer.

Each synthetic scenario has known evidence, policy, action, and outcome expectations.



EIGHT SCORED DIMENSIONS



104

synthetic scenarios
16 risk + 88 trajectories

8 workflows · 11 conditions

Text · chat · voice · image · tool failure · prompt injection

104 / 104 passed

All eight dimensions = 100%

Hard gate

One policy or safety violation fails CI.

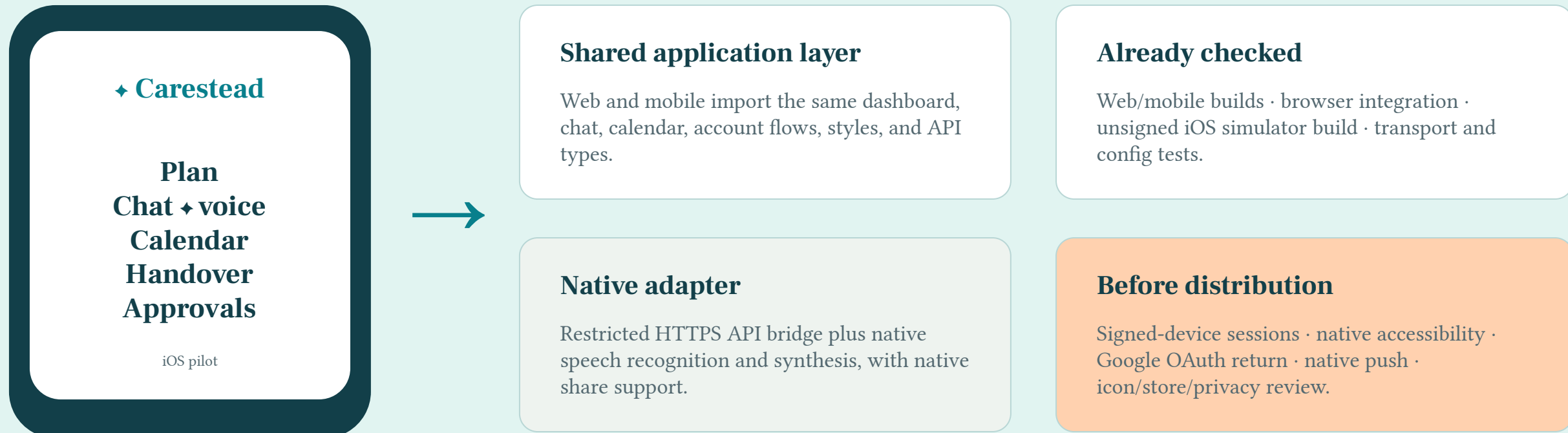
Synthetic regression benchmark

Not clinical validation or measured caregiver impact.

MOBILE

The same care workspace is moving to iPhone.

A Capacitor/iOS pilot reuses the responsive Carestead UI, shared types, and the same authenticated backend APIs.



STATUS · Integration in progress. Pilot, not an App Store release.

REAL-WORLD IMPACT

A scattered care story becomes a safe, shared next step.

Before Carestead

Fragmented information
Repeated reconstruction
Unclear responsibility ownership
Follow-up and handover gaps

With Carestead

Shared recipient context
Named ownership and reviewable actions
Visible outcomes and audit history
Faster, clearer caregiver handover

PILOT MEASURES

Coordination time · unresolved responsibilities · unnecessary alerts · handover completeness · caregiver confidence

Less to carry. More care to give.